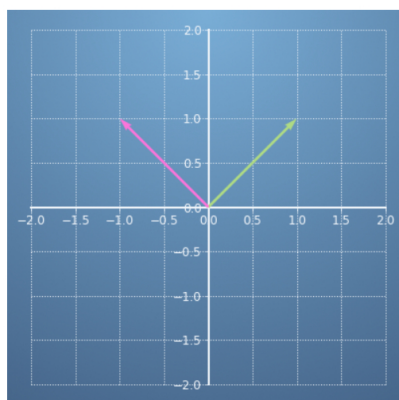


1 point

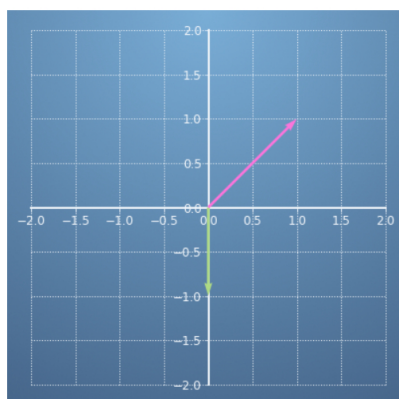


1. Compute the angle between $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ using the inner product defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} \mathbf{y}$$

- ☐ 0.35 rad (20°)
- ☒ 1.2 rad (69°)
- ☐ 1.57 rad (90°)

1 point



2. Compute the angle between $\mathbf{x} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ using the inner product defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 5 \end{bmatrix} \mathbf{y}.$$

To aid in computing this angle and the next ones in this quiz, let's write an expression in Python for the angle between two vectors using a non-standard inner product.

$$\text{Remember } \cos \alpha = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} \sqrt{\langle \mathbf{y}, \mathbf{y} \rangle}}$$

Complete the expressions for norm_x and norm_y and then run the code. You might find the NumPy function `np.sqrt` [useful](#).

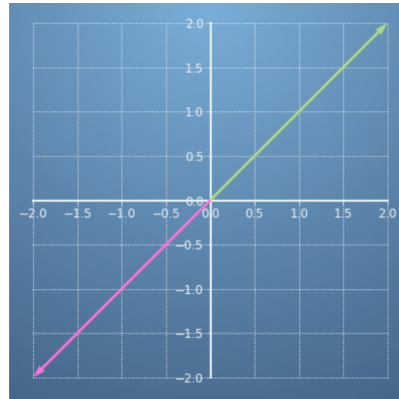
```
1 # the matrix A defines the inner product
2 A = np.array([[1, -1/2], [-1/2, 5]])
3 x = np.array([0, -1])
4 y = np.array([1, 1])
5
6 def find_angle(A, x, y):
7     """Compute the angle"""
8     inner_prod = x.T @ A @ y
9     # Fill in the expression for norm_x and norm_y below
10    norm_x = np.sqrt(x.T @ A @ x)
11    norm_y = np.sqrt(y.T @ A @ y)
12    alpha = inner_prod / (norm_x * norm_y)
13    angle = np.arccos(alpha)
14    return np.round(angle, 2)
15
16 find_angle(A, x, y)
```

Run

Reset

- ☐ 2.69 rad (154°)
☐ -0.9 rad (-52°)
☒ 2.69 rad (154°)

1 point

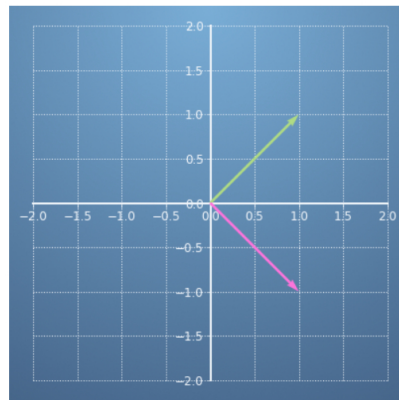


3. Compute the angle between $\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ using the inner product defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \mathbf{y}$$

- ☐ 0 rad (0°)
☒ 3.14 rad (180°)

1 point



4. Compute the angle between $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ using the inner product defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \mathbf{y}$$

```

1 # Fill in the arrays and use the function `find_angle` defined for you to aid in you
2 A = np.array([[1, 0], [0, 5]])
3 x = np.array([1, 1])
4 y = np.array([1, -1])
5
6 find_angle(A, x, y)

```

Run
Reset

- ☐ -2.3 rad (-131°)
☐ 1.57 rad (90°)
☒ 2.3 rad (131°)
☐ -1.57 rad (-90°)

5. Compute the angle between $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ using the inner product defined by

1 point

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} \mathbf{y}$$

```

1 # Fill in the following arrays and use `find_angle` to aim your calculation.
2 A = np.array([[1, 0, 0],[0, 2, -1],[0, -1, 3]])
3 x = np.array([1, 1, 1])
4 y = np.array([2, -1, 0])
5
6 find_angle(A, x, y)

```

Run

Reset

- ☐ 1.31 rad (75°)
- ☐ 0.2 rad (11°)
- ☒ 1.37 rad (78°)